



VERA

ABSTRACT

WELCOME TO VERA ...

WHERE WE HELP YOU FIND WHAT'S TRENDING IN YOUR CLOSET. Our project refines the fashion industry through a dual-filter recommendation system, inspired by the success of AI-driven retailers like Stitch Fix. The first component utilizes content-based filtering, built on our object detection and image classification engine. This engine ideally analyzes user interactions—such as likes or dislikes—to tailor fashion suggestions directly aligned with individual tastes¹. By incorporating user feedback from an initial quick survey upon signup, the system effectively navigates the common cold start problem, providing personalized recommendations.

The second component of our system, collaborative filtering, enhances the precision of recommendations by analyzing fashion trends. This approach synthesizes data from current fashion trends and user preference patterns to refine and optimize the suggestions provided by the content-based filter. Currently, our prototype uses a simulated dataset focused on color preferences, but future expansions will include web-scraped real data to capture a broader range of fashion attributes directly from user interactions.

The intended impact of our project is to enable an average user with relevant, accessible, and personalized fashion recommendations. By merging daily outfit suggestions with current fashion trends and the users' existing wardrobes, our platform encourages sustainable fashion practices. This approach not only reduces unnecessary consumption and waste but also helps users rediscover and optimize their current wardrobe pieces. We aim to create a culturally conscious fashion platform that not only keeps pace with the latest trends but also promotes responsible fashion consumption, aligning with Vera's modern values of sustainability and personalization.

DATA

Our initial model utilized in preliminary EDA was discarded in favor of the model discussed in this report. The original model was trained on a dataset of 45 thousand images, pulled from the fashion site "Myntra." Originally selected due to the convenience of consistently centered, white background images, we opted to retrain our model on data that is more relevant to American fashion and includes greater clothing diversity. The goal for our model is that it may detect fashion attributes on unfamiliar clothing images with relatively high accuracy, which our original model struggled to accomplish given the difference in fashion present on Myntra vs. New York Fashion Week (NYFW) photos.

¹ See Exhibit A

The new dataset is designed for the task of fine-grained image classification of fashion items. These images contain better diversity in models, display of clothing, and background color.² It comprises over one million fashion images sourced from the Wish platform. These images are divided into three sets: 1,012,947 training images, 9,897 validation images, and 39,706 test images. The dataset is structured into 228 fine-grained fashion attribute-level classes, which are further grouped into 8 high-level fashion groups professionally defined by the fashion industry. These attributes are categorized into several high-level fashion groups, and each group contains multiple fine-grained attributes.

For example, one of the groups is "Gender," which includes attributes like men's, women's, or unisex clothing. Another group could be "Apparel," with more detailed attributes like dresses, shirts, pants, etc. Further groups include "Color," "Material," "Pattern," and so forth, each subdivided into even more specific categories, such as "cotton," "silk," "leather" for materials, or "striped," "polka dot," "plaid" for patterns.

For the collaborative filtering and color analysis portion of our dataset, we created a simulated dataset for testing purposes. While the dataset holds little usefulness in real-life application, it allowed us to review how the collaborative filtering methodology would theoretically work. User aspects like gender, age, state of residency (limited to the U.S.) and personality traits were assigned to one color preference and one-color dislike.

For the fashion trend analysis data, we pulled from Vogue and Now Fashion, an online platform that serves as an archive for fashion show images worldwide. It offers high quality photo collections from runway featuring both iconic designers and emerging talents. The site is continually updated to include the latest collections, especially during fashion week events in major cities like Paris, Milan, London, and New York. This ensures that fashion enthusiasts (such as ourselves!) can stay up-to-date with the latest trends and styles showcased on the runways.

METHODOLOGY

IMAGE CLASSIFICATION

1. Data Preprocessing

First, we assembled a consolidated dictionary linking each image ID with its corresponding URL and label ID, facilitating cross-referencing between the image's tagging data and its processing details. Next, we load the image from its given file path. The image is then converted from its original JPEG format into a tensor, which is a multi-dimensional array that accurately depicts the image in a three-channel RGB color space. The next step involves standardizing the image size to a uniform 128x128 pixels. This dimension is usually expected by various neural networks for image analysis.³ To finalize the preprocessing, the function adjusts the image's pixel intensity values, scaling them from their original span of 0 to 255 to a normalized range between 0 and 1. This normalization aids in neural network's ability to learn during the training phase.

² See Exhibit B

³ Best way to resize images for object recognition. (2023, January 28). TensorFlow Forum. <https://discuss.tensorflow.org/t/best-way-to-resize-images-for-object-recognition/14504>

2. Model Considerations

Our image classification goal, in this context, is aimed at categorizing each image or detected object within an image into predefined classes, which in the case of the fashion dataset, include various types of apparel and accessories.

To achieve this, we utilized the TensorFlow framework, which supports the deployment of deep learning algorithms. The model used for object detection is based on the Convolutional Neural Network (CNN) architecture, which is suited for image processing.⁴ CNNs are designed to adaptively learn spatial hierarchies of features through backpropagation. By using layers of convolutions with learnable filters followed by pooling layers, these networks can process raw image pixels and learn to identify patterns that correspond to different objects in the images.

The object detection process itself is facilitated by adjusting the pre-trained model to better fit the specific needs of fashion item detection. The model's later layers are fine-tuned and retrained on the new dataset to adapt its learning to the particular features of fashion items.

Our other motivation for selecting TensorFlow is its real-time object detection ability, which can be integrated into applications for immediate analysis of fashion images. This capability supports our thesis of real-time fashion feedback and support by potential Vera users. This way, users may upload images and receive instant classifications or tagging on their clothing items.

3. Model Implementation

The model begins with a sequential constructor, stacking layers to process the input images. It starts with a two-dimensional convolutional layer equipped with 32 filters of 3x3 size, employing the ReLU activation function. This layer is prepared to handle input images with a resolution of 128x128 pixels in a three-channel format for RGB images.

Following the initial convolutional layer is a max pooling layer that serves to decrease the spatial dimensions by a factor of two, effectively downsizing the feature maps while retaining the most critical elements. Another convolutional layer follows, this time with 64 filters to allow the detection of more complex features. A subsequent max pooling layer further downsizes the feature maps, continuing the pattern of convolution and pooling that is characteristic of CNNs.

Once the spatial features have been condensed, a flatten layer transitions the data from a two-dimensional form to a one-dimensional vector, making it suitable for processing by standard fully connected layers. Next, a dense layer with 512 neurons applies the ReLU activation function to facilitate the network's ability to learn non-linear relationships. To mitigate the risk of overfitting, a dropout layer randomly nullifies 50% of the neurons during training.

The final layer of the network is another dense layer that corresponds to the number of classes represented in the multi-label binarizer, utilizing the Sigmoid activation function. This allows the model to predict the probability of each class independently, essential for multi-label classification tasks where each image could be tagged with multiple labels.

The model is compiled with the Adam optimization algorithm and the binary cross-entropy loss function, helping with Vera's multi-label classification problems. The 'accuracy' metric is tracked, giving insight into the performance

⁴ (5) Understanding convolutional neural networks (CNNs): Best for image classification | LinkedIn. (2023, June 25). <https://www.linkedin.com/pulse/understanding-convolutional-neural-networks-cnns-best-al-ameen/>

of the model during training. The model architecture is thus fully prepared to undergo training for tasks that require the identification of multiple labels within a single image.

4. Test

We tested the fitted model on images outside the validation dataset. In honor of our fashion choices, Natalie posed in her PJs and robe to test the model. The image classifier and object detection portions of the model were able to properly identify two labels: Long sleeves and round neck⁵.

COLOR ANALYSIS

1. ANALYSIS

Our colorCoder utilized Python packages such as Numpy for data manipulation, Matplotlib for visualization, and OpenCV for image processing are used throughout the color analysis. OpenCV hosts image operation capabilities including reading, resizing, and converting color spaces, for preparing images for analysis.

Utilizing the Scikit-learn library's K-Means clustering to segment image colors into distinct groups, we identify the most common color clusters.⁶ This method determines the centroid of pixel groups, summarizing the most dominant colors in the image. Adjusting the number of clusters allows for finer control over the granularity of the color grouping, which may enhance precision at the detriment of processing time.

The output of our colorCoder is a “palette” that displays the most common colors, each representing a cluster derived from the K-Means algorithm. These results are shown in proportion to their occurrence, such that dominant colors with greater frequency within the image are given a larger portion of the palette.

2. COMPARING IMAGES

To analyze the similarity of colors, we had to address the limitations of traditional RGB color space and opt for the CIELAB color space for its superior accuracy in mimicking human color perception.⁷ RGB, while common, does not align with how our eyes discern color differences, leading to misleading similarity assessments. CIELAB includes dimensions for luminance (L*) and color channels (a, b)—specifically designed to parallel human vision. This outputs more precise and perceptually relevant measurements.

Once images were correctly translated to relevant metrics, we performed simple Euclidean distance on the three attributes within the orthogonal color space. Given that assessing the “similarity” of color is subjective to the human eye, we tested with the Fall Winter 2024 Chanel line, with Natalie’s favorite look from the line. This was met with arguable success given the image pool of 6.⁸

3. “YOU HAVE SUCH A COLORFUL PERSONALITY!”

The study titled "Exploring the relationships between personality and color preferences" by Juliet Jue and Jung Hee Ha investigates how different personality types are associated with color preferences. Participants were asked to associate colors with personality-descriptive words, complete the Ten Item Personality Inventory (TIPI), and rank

⁵ Exhibit C

⁶ A, M. R. L. (2021, December 16). Finding the most common colors in Python - towards data science. Medium. <https://towardsdatascience.com/finding-most-common-colors-in-python-47ea0767a06>

⁷ Antoniadis, P., & Antoniadis, P. (2024, March 18). How to compute the similarity of colours | Baeldung on Computer Science. Baeldung on Computer Science. <https://www.baeldung.com/cs/compute-similarity-of-colours>

⁸ Exhibit D

their color preferences. The results demonstrated distinct color associations for all five personality types, for example, people who scored high on agreeableness tended to prefer yellow, a color typically associated with warmth and friendliness.⁹

The study further explored these relationships through cluster and regression analyses, revealing that individuals' color preferences could predict their personality characteristics. The findings suggested that color preferences could reflect deeper psychological profiles. At Vera, we aim to understand how the interplay of fashion, color, and aesthetic contributes to one's personal expression. By creating a faux dataset based on the principles outlined in this study, we aim to combine user personality with their experience with color. Like how Spotify provides a personalized DayList to its user, the goal would be to ultimately provide a "ColorList" with your color based on your personality.

TREND ANALYSIS

1. SPRING 2024 TRENDS

Our image classification and object detection model was applied major fashion house lines of Louis Vuitton, Acne Studios, and Chanel to determine current trends. After applying the model with TensorFlow Vectorization, collecting the predicted label codes, and printing them out in readable format, our top 5 recommendations were: backless dresses, drawstring pants, baggy jeans, halter tops, and the color gray.¹⁰

2. COLOR TRENDS, PERSONAL COLOR CHOICES

We applied our colorCoder tool to Vogue magazines from the year 2023 to identify trending colors. This method, while focused primarily on Vogue's most recent collections and fall fashion lines from various brands, could ideally extend to a comprehensive analysis across all major fashion houses. Our findings highlight a general preference for neutral colors throughout the year, with a noticeable increase in pink during the summer months as reported by Vogue. The fall and winter collections predominantly feature cool tones and neutrals.

Additionally, we successfully tested the colorCoder on Pinterest pins. Natalie manually downloaded images from her clothing inspiration board on Pinterest, a process that could be streamlined through future API integration. The analysis from this exercise pinpointed a preference for blues and neutrals, with occasional inclinations towards green in Natalie's selections¹¹. This practical application underscores the tool's potential for personalizing color preferences and insights into current fashion trends.

FUTURE WORK

Our object detection and image classification model are operational and effective across various backgrounds and lighting conditions, and its tagging capabilities are satisfactory. After proving functionality to output a broader spectrum of attributes for different trend cycles, from Fall to Spring 2024, we feel confident in the model's viability with the current training and tuning.

Regarding our color analysis, the preliminary dataset we created does not fully represent the scale and scope needed for comprehensive analysis. Our intention is to augment this dataset either by aggregating data from online sources

⁹ Jue, J., & Ha, J. H. (2022). Exploring the relationships between personality and color preferences. *Frontiers in psychology*, 13, 1065372. <https://doi.org/10.3389/fpsyg.2022.1065372>

¹⁰ Exhibit F

¹¹ Exhibit E

through web scraping or by incorporating findings from existing research that links color preferences with demographic data. This would provide a more relevant and accurate framework for comparability.

The next phase involves integrating the object detection and color analysis segments. We're exploring the most effective means to synchronize these aspects and refine the similarity metrics we use. Anaya, who is so talented at front-end development¹², is already designing a user interface with HTML and CSS that aligns with the prototype shown in Exhibit A. Our ultimate goal is to launch a web application that personalizes fashion recommendations and trends based on user's preferences revealed through a personality quiz, addressing the initial engagement, or "cold start," issue. Then, we can fully realize the content-based recommendation.

CONCLUSION

Vera aims to apply a dual-filter recommendation system that merges content-based and collaborative filtering to tailor fashion suggestions. The content-based component utilizes an image classification engine to analyze user interactions, like preferences and dislikes, to provide personalized fashion suggestions right from the start. Collaborative filtering complements this by incorporating data on current fashion trends, sourced from platforms like Vogue and Now Fashion, to further refine and enhance these recommendations.

This system aims to provide recommendations that not only match current fashion trends but also encourage sustainable fashion practices. By helping users utilize their existing wardrobes more effectively, Vera seeks to reduce wasteful consumption and promote sustainability in fashion choices.

We enjoyed working on this project together to address an issue we all felt was relevant to our daily life; trying to balance self-expression and connection with the fashion community against the consumerists cycle enabled by the larger fashion industry. The learning threshold for this project was high, considering we did not have prior experience in image analysis of any kind. Inspired by the case studies in class, and enabled by an understanding of how this application can flourish into a dual recommendation engine (kudos to the hybrid recommendation engine assignment) we felt capable. With a functioning image classification and object detection model that operates successfully on unfamiliar datasets, and a collaborative filtering framework based on color analysis, we now turn our focus on building out the front-end for a functioning content-based recommendation system.

Everyone at Vera is genuinely enthusiastic and passionate about this project, driven by our academic insights from this class and self-taught knowledge. We're excited to keep building on this momentum as we move the project forward into the summer!

¹² Source: Natalie Hammel

The Vera Team



Natalie Hammel
co-Founder



Ananya Shah
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Thank you to Deba for continuous support and guidance during MKTG4604!

APPENDIX

GITHUB REPO: [link](#).

EXHIBIT A: PROTOTYPE ADDRESSING COLD-START PROBLEM

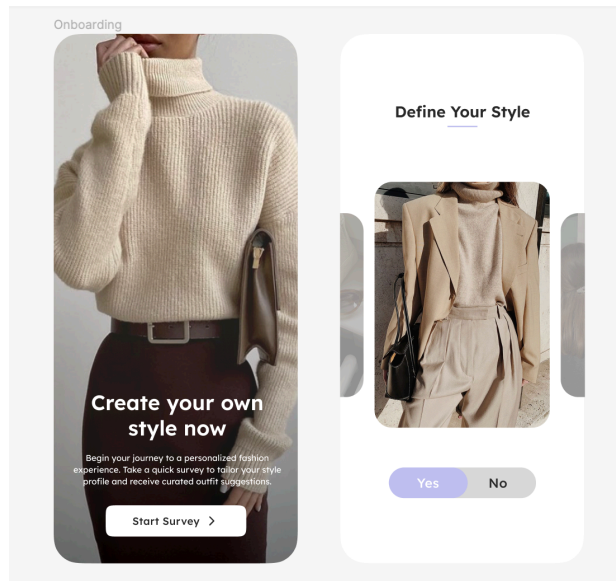


EXHIBIT A: COMPARING MYNTRA (LEFT) WITH WISH (RIGHT)



EXHIBIT B: COLORS AND ASSOCIATED PERSONALITY TRAITS

Table 1

Colors associated with personality words.

Personality type	Adjective	First ranking (%)	Second ranking (%)	Third ranking (%)
Extraversion	Extraverted (+)	Red (24.7)	Yellow (19.3)	Orange (12.8)
	Enthusiastic (+)	Red (43.2)	Orange (16.4)	Yellow (11.1)
	Reserved (–)	White (16.2)	Dark blue (15.2)	Black (14.5)
	Quiet (–)	White (22.6)	Gray (15.4)	Green (14.8)
Agreeableness	Sympathetic (+)	Yellow (24.3)	Green (19.4)	Orange (16.6)
	Warm (+)	Yellow (27.2)	Orange (24.6)	Red (12.8)
	Critical (–)	Black (21.3)	Gray (20.0)	Dark blue (13.2)
	Quarrelsome (–)	Red (32.5)	Black (13.6)	Purple (9.5)
Conscientiousness	Dependable (+)	Green (25.0)	Light blue (21.6)	Dark blue (16.2)
	Self-disciplined (+)	Green (17.7)	White (17.5)	Dark blue (15.7)
	Disorganized (–)	Purple (22.7)	Gray (17.5)	Red (13.7)
	Careless (–)	Red (22.0)	Orange (12.1)	Purple (11.7)
Emotional stability	Calm (+)	Green (18.0)	White (16.6)	Dark blue (15.1)
	Emotionally stable (+)	Green (24.6)	Light blue (15.6)	White (13.9)
	Anxious (–)	Purple (21.7)	Red (17.9)	Gray (14.5)
	Easily upset (–)	Red (44.0)	Orange (12.5)	Yellow (8.9)
Openness to new experiences	Open to new exp. (+)	Green (19.2)	Yellow (19.0)	Light blue (18.8)
	Complex (+)	Purple (28.5)	Gray (15.0)	Dark blue (11.1)
	Conventional (–)	Black (30.6)	Gray (18.0)	White (11.8)
	Uncreative (–)	Black (31.0)	Gray (22.0)	White (16.2)

The + symbol shows the core of that personality, and the – symbol shows the opposite trait of that personality.

EXHIBIT C: TESTING WITH NATALIE'S "OUTFIT"



```
human_readable_labels = translate_labels(predicted_1
print(f"Predicted Labels: {human_readable_labels}")
```

Predicted Labels: ['Long Sleeved', 'Round Neck']

EXHIBIT D: COMPARING “SIMILAR” COLORS



EXHIBIT E: COLOR ANALYSIS OF PINTEREST PHOTOS

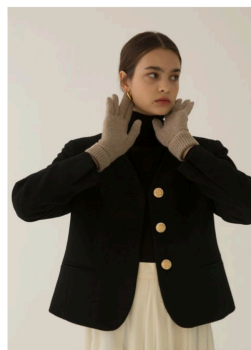


EXHIBIT F: TRENDS FOR SPRING 2024

